GRADE 7 ACCELERATED · UNIT 5 · LESSON 1

Scale Drawings

Question Bank 5.01 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the scale factor from the original to the copy.

1. A 4 cm side becomes 12 cm.

k = 3

2. A 9 cm side becomes 3 cm.

k = ⅓

3. A 2 cm by 7 cm rectangle is copied at 5 cm by 17.5 cm.

k = 2.5

5 ÷ 2 and 17.5 ÷ 7 both give 2.5.

4. A 16 cm side becomes 12 cm.

k = ¾

For questions 5 through 8, use the scale factor or the scale.

5. The scale factor is 4 and a side of the original is 2.5 cm. How long is that side in the copy?

10 cm

6. The scale factor is 0.5 and a side of the original is 9 cm. How long is that side in the copy?

4.5 cm

7. A map's scale is $1\,\text{cm} : 30\,\text{km}$. Two towns are 5.5 cm apart on the map. How far apart are they?

165 km

8. On the same map, a lake is 45 km long. How long is it on the map?

1.5 cm

Back to the map: divide by 30.

Choose one.

9. A drawing has a 65° angle. The real figure is a scale copy at $k = 3$. What is the angle in the real figure?

- ○ A. 68°
- ○ B. 195°
- ● **C. 65°**
- ○ D. 21.7°

C

A scale factor multiplies lengths; angles do not change.